

DATA PORTFOLIO: ROAD ACCIDENT ANALYSIS & RISK MODELING

Author: Data Analyst & Business Intelligence Specialist | **Dataset Focus:** Multi-Year Traffic Casualties (2021-2022)

1. Project Overview & Business Case

This data analytics initiative transforms raw, high-volume traffic incident records into an interactive, high-performance executive dashboard suite. The project models historical road accidents across distinct environmental vectors, infrastructure variations, and localized geographic profiles. The primary operational objective is delivering data-driven spatial and situational intelligence to help urban planning stakeholders deploy high-ROI, targeted road safety countermeasures.

2. Core Performance Metrics & KPIs

Metric Tier	Observed Metrics & Trends (2021-2022)
Total Volume Metrics	195,737 Total Casualties from **144,419 Core Traffic Accidents** , representing an optimized YoY reduction of approximately 11.7%.
Severity Stratification	• Fatal Casualties: 2,855 (Down 26.40% YoY) • Serious Casualties: 27,045 (Down 16.30% YoY) • Slight Casualties: 165,837 (Down 10.82% YoY)
Vulnerability Breakdown	Passenger Cars represent the primary operational liability, encompassing over **80% of total casualties** (155,804 incidents). Vans follow as a secondary risk vector.

3. Strategic Data Insights

- **Structural Road Infrastructure Risks:** **Single Carriageways** stand out as the highest-risk configurations by a wide margin, causing over **144K+ casualties**. This highlights a clear need for infrastructure updates like physical medians.
- **Urban Location Multipliers:** Casualty volumes in **Urban Environments** drastically outpace Rural areas (~62% vs 38%). This demonstrates that high traffic density and complex intersections significantly amplify pedestrian and commuter incident risks.
- **Environmental Factor Analysis:** Counter-intuitively, more than ****73.8%** of incidents occur during peak Daylight hours** under **Fine Weather Conditions (80.7%)**. This strongly indicates that driver distraction, tailgating, and high-volume rush-hour congestion outweigh nighttime visibility issues.

4. Data Engineering & Optimization

Architecture & Size Compression: To make deployment efficient, the system pipeline was engineered to handle large-scale database rows while staying under a strict **25 MB footprint**. This was achieved by: removing unused fields via Power Query, transforming high-cardinality DateTime values into decoupled dimensional dimensions, and disabling redundant global auto-date features. The resulting optimized schema ensures exceptionally fast visualization loads and zero-latency cross-filtering across dynamic maps and trend charts.

5. Actionable Recommendations for HR / Stakeholders

1. **Data-Driven Enforcement:** Deploy automated speed cameras and active warning indicators along high-risk **Single Carriageways** to immediately curb high-speed accidents.
2. **Urban Safety Upgrades:** Redesign high-incident urban intersections, adding protected bike lanes and safer pedestrian zones to address the high rate of urban casualties.
3. **Public Awareness:** Launch driver awareness programs focused on passenger car safety during peak daylight rush hours, targeting the main causes of driver distraction.

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